

Smart Hydration Tracking Water Bottle

ABSTRACT

A water bottle apparatus includes a fluid reservoir, a capacitive fill-level sensor, a low-power wireless transceiver, and a companion mobile application. The sensor periodically measures the remaining fluid volume and transmits hydration events to a paired device, enabling the application to compute daily intake and issue reminder notifications when intake falls below a user-configured threshold.

BACKGROUND

Existing hydration reminder devices typically rely on fixed time-interval alerts that disregard actual fluid consumption, producing reminders even when a user has already consumed an adequate volume of water. There is a need for a bottle that measures actual consumption directly and adapts its reminder schedule accordingly.

SUMMARY

In one embodiment, the bottle housing defines an internal cavity fitted with a capacitive sensor array along its vertical axis. A microcontroller samples the array at a fixed interval, computes an estimated fluid level, and stores a rolling log of consumption events in local memory. When paired with a mobile device over Bluetooth Low Energy, the bottle synchronizes this log and receives an updated reminder schedule computed from the user's historical intake pattern.

CLAIMS

1. A hydration tracking apparatus comprising: a fluid reservoir; a capacitive sensor array disposed along an interior wall of the reservoir; a microcontroller coupled to the sensor array and configured to compute a fluid level from sensor readings; and a wireless transceiver configured to transmit consumption events to a paired mobile device. 2. The apparatus of claim 1, wherein the microcontroller further computes a rolling daily consumption total and triggers a local indicator when the total falls below a configured threshold. 3. The apparatus of claim 1, wherein the wireless transceiver operates according to a Bluetooth Low Energy protocol.